

Duality closes the intermediate-trace gap for positive isometries of $L^1 \cap L^\infty$

Candidate full solution packet for arXiv:2007.02324

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Status. Candidate full solution, likely valid, pending expert review. The argument answers the open problem in Remark 5.6 of de Jager–Conradie [1] and in fact proves a slightly stronger theorem.

1 Source question

The source is Pierre de Jager and Jurie Conradie, *Extreme point methods in the study of isometries on certain non-commutative spaces*, arXiv:2007.02324, Remark 5.6 on printed/PDF page 21 [1]. The complete source passage is reproduced below directly from the paper.

that $U(x) = a\Phi(x)$ for every $x \in L^1 \cap L^\infty(\mathcal{A})$. In both cases it would then be desirable to refine the initial representation.

Remark 5.6. We make a few brief comments regarding the assumption $2 \leq \tau(\mathbf{1}), \nu(\mathbf{1}) \leq \infty$. Note that if $\tau(\mathbf{1}_{\mathcal{A}}) < \infty$, then $L^1 \cap L^\infty(\mathcal{A}) = \mathcal{A}$. If, in addition, $\tau(\mathbf{1}_{\mathcal{A}}) \leq 1$, then for any $x \in L^1 \cap L^\infty(\mathcal{A}) = \mathcal{A}$, we have that $\|x\|_1 = \|x\mathbf{1}\|_1 \leq \|x\|_\infty \tau(\mathbf{1}) \leq \|x\|_\infty$ and therefore $\|x\|_{1 \cap \infty} = \max\{\|x\|_1, \|x\|_\infty\} = \|x\|_\infty$. Consequently, if $\tau(\mathbf{1}_{\mathcal{A}}), \nu(\mathbf{1}_{\mathcal{B}}) \leq 1$, then surjective isometries from $L^1 \cap L^\infty(\mathcal{A})$ onto $L^1 \cap L^\infty(\mathcal{B})$ are described by Kadison’s Theorem ([24, Theorem 7]). To show that U is projection disjointness preserving in the proof of Lemma 5.2 we use a technique which relies on being able to use two orthogonal projections p and q for which we know the precise values of $\|U(p)\|_1$ and $\|U(q)\|_1$ in order to obtain equality in Clarkson’s inequality (see (5.1)). For $\tau(p) = 1 = \tau(q)$ we have access to the characterization of the extreme points of the unit ball, but not otherwise. For this reason we have assumed that $\tau(\mathbf{1}_{\mathcal{A}}) \geq \tau(p+q) \geq 2$. This also plays a role in showing that U maps projections to projections and since we later apply this to U^{-1} , we also assume that $\nu(\mathbf{1}_{\mathcal{B}}) \geq 2$. It is therefore an open problem to describe positive surjective isometries from $L^1 \cap L^\infty(\mathcal{A})$ onto $L^1 \cap L^\infty(\mathcal{B})$ if $1 < \tau(\mathbf{1}_{\mathcal{A}}) < 2$ or $1 < \nu(\mathbf{1}_{\mathcal{B}}) < 2$.

Transcription of the question. Describe the positive surjective isometries

$$U : L^1 \cap L^\infty(\mathcal{A}) \longrightarrow L^1 \cap L^\infty(\mathcal{B})$$

if $1 < \tau(\mathbf{1}_{\mathcal{A}}) < 2$ or $1 < \nu(\mathbf{1}_{\mathcal{B}}) < 2$. Here $(\mathcal{A}, \tau)$ and $(\mathcal{B}, \nu)$ are non-atomic semifinite von Neumann algebras, as throughout the source section.

For a semifinite traced algebra $(\mathcal{M}, \mu)$, write

$$E_{\mathcal{M}} = L^1(\mathcal{M}, \mu) \cap \mathcal{M}, \quad \|x\|_{1 \cap \infty} = \max\{\|x\|_1, \|x\|_{\infty}\},$$

and

$$F_{\mathcal{M}} = L^1(\mathcal{M}, \mu) + \mathcal{M}, \quad \|a\|_{1+\infty} = \inf_{a=a_1+a_{\infty}} (\|a_1\|_1 + \|a_{\infty}\|_{\infty}).$$

2 Proof intuition

The obstruction noted by the source is specific to its direct extreme-point argument: when the total trace is below two, there are not two orthogonal trace-one projections. The key idea is to change sides of the duality rather than repair that argument.

If $1 < \tau(\mathbf{1}_{\mathcal{A}}) < 2$, then $E_{\mathcal{A}} = \mathcal{A}$ and $\mathbf{1}_{\mathcal{A}}$ is an order unit. Since a positive onto isometry is an order isomorphism, its image is an order unit in $E_{\mathcal{B}}$. An infinite-trace intersection space has no order unit; hence $\nu(\mathbf{1}_{\mathcal{B}}) < \infty$ and $E_{\mathcal{B}} = \mathcal{B}$. Extreme points then rule out $\nu(\mathbf{1}_{\mathcal{B}}) \leq 1$.

Both algebras are therefore finite with total trace greater than one. The normal preadjoint of U is a positive surjective isometry for the polar $L^1 + L^{\infty}$ norms. The source paper has already classified precisely those sum-space isometries when both total traces exceed one. It gives a trace-preserving Jordan isomorphism, and the trace pairing shows that U is its inverse.

3 Result

Theorem 1 (Intermediate-trace classification). *Let $(\mathcal{A}, \tau)$ and $(\mathcal{B}, \nu)$ be non-atomic semifinite von Neumann algebras, and let*

$$U : E_{\mathcal{A}} \longrightarrow E_{\mathcal{B}}$$

be a positive surjective linear isometry. Suppose that $1 < \tau(\mathbf{1}_{\mathcal{A}}) < \infty$ or $1 < \nu(\mathbf{1}_{\mathcal{B}}) < \infty$. Then both identity traces are finite and greater than one, they are equal, and U is a trace-preserving Jordan $$ -isomorphism from $\mathcal{A}$ onto $\mathcal{B}$.*

Conversely, every trace-preserving Jordan $$ -isomorphism from $\mathcal{A}$ onto $\mathcal{B}$ restricts to a positive surjective isometry $E_{\mathcal{A}} \rightarrow E_{\mathcal{B}}$. In particular, this completely answers the source question, whose finite trace is assumed to lie in $(1, 2)$.*

Proof. De Jager–Conradie prove that a positive isometry between noncommutative symmetric spaces reflects positivity; if it is onto, it and its inverse are order isomorphisms and are normal [2, Lemma 3.1]. Thus, if the finite trace in the hypothesis is $\nu(\mathbf{1}_{\mathcal{B}})$, we may replace U by U^{-1} . Assume from now on that

$$1 < t := \tau(\mathbf{1}_{\mathcal{A}}) < \infty.$$

Then $E_{\mathcal{A}} = \mathcal{A}$ as a set, and $\mathbf{1}_{\mathcal{A}}$ is an order unit.

Step 1: the target trace is finite. Set $y = U(\mathbf{1}_{\mathcal{A}}) \geq 0$. For every $z \in (E_{\mathcal{B}})_{\text{sa}}$, positivity of U and U^{-1} gives

$$-\|U^{-1}z\|_{\infty}y \leq z \leq \|U^{-1}z\|_{\infty}y.$$

Hence y is an order unit of $E_{\mathcal{B}}$.

We claim that $E_{\mathcal{B}}$ has no order unit when $\nu(\mathbf{1}_{\mathcal{B}}) = \infty$. Indeed, let $0 \leq y \in E_{\mathcal{B}}$. For each integer $k \geq 1$, Markov's inequality gives

$$\nu(\mathbf{1}_{(1/k, \infty)}(y)) \leq k\nu(y) < \infty,$$

so $e_k = \mathbf{1}_{[0, 1/k]}(y)$ has infinite trace. By semifiniteness and non-atomicity, recursively choose nonzero, mutually orthogonal projections $q_k \leq e_k$ with $\nu(q_k) \leq 2^{-k}$. The strong sum $q = \sum_{k \geq 1} q_k$ is a projection of finite trace, so $q \in E_{\mathcal{B}}$. If $q \leq cy$ for some $c > 0$, compression by q_k yields

$$q_k \leq cq_k y q_k \leq \frac{c}{k} q_k,$$

which is impossible for $k > c$. Thus y is not an order unit. This contradiction proves $s := \nu(\mathbf{1}_{\mathcal{B}}) < \infty$, and hence $E_{\mathcal{B}} = \mathcal{B}$.

Step 2: the target total is greater than one. The source paper characterizes the extreme points of the intersection-space unit ball: they are the partial isometries v satisfying

$$\nu(|v|) = \min\{1, \nu(\mathbf{1}_{\mathcal{B}})\}$$

(and analogously for $\mathcal{A}$) [1, Proposition 5.1]. Since $\mathcal{A}$ is non-atomic and $t > 1$, it has two distinct projections p_1, p_2 of trace one. (Starting from one such projection, move equal small-trace pieces between it and its nonzero complement.) Both p_i are extreme points.

A surjective linear isometry preserves extreme points, so each $U(p_i)$ is a positive extreme point. A positive partial isometry is a projection. If $s \leq 1$, the displayed characterization forces $\nu(U(p_i)) = s = \nu(\mathbf{1}_{\mathcal{B}})$, whence faithfulness gives $U(p_i) = \mathbf{1}_{\mathcal{B}}$. This contradicts injectivity because $p_1 \neq p_2$. Therefore $s > 1$.

Step 3: pass to the normal dual. On each finite algebra the intersection norm is equivalent to the operator norm:

$$\|x\|_{\infty} \leq \|x\|_{1 \cap \infty} \leq \max\{1, \mu(\mathbf{1})\} \|x\|_{\infty}.$$

Consequently the positive normal maps $U : \mathcal{A} \rightarrow \mathcal{B}$ and U^{-1} have inverse normal preadjoints. Write

$$V : L^1(\mathcal{B}, \nu) \longrightarrow L^1(\mathcal{A}, \tau), \quad \tau(V(b)x) = \nu(bU(x)). \quad (1)$$

For normal functionals, intersection/sum polarity gives

$$\|a\|_{1+\infty} = \sup\{|\mu(ax)| : x \in \mathcal{M}, \|x\|_{1 \cap \infty} \leq 1\}. \quad (2)$$

This is the standard noncommutative Köthe-duality formula [3]. Equivalently, it is the polar formula for the maximum norm on an intersection: the polar unit ball is the closed convex hull of the two polar unit balls, equivalently the infimal-convolution sum norm. It remains valid here after restricting to normal functionals, which are represented by L^1 .

Using (1), (2), and $U(B_{E_{\mathcal{A}}}) = B_{E_{\mathcal{B}}}$, we obtain

$$\|V(b)\|_{1+\infty} = \sup_{\|x\|_{1 \cap \infty} \leq 1} |\nu(bU(x))| = \sup_{\|z\|_{1 \cap \infty} \leq 1} |\nu(bz)| = \|b\|_{1+\infty}.$$

The preadjoint of U^{-1} is V^{-1} , and positivity of U and U^{-1} implies positivity of both preadjoints. Thus

$$V : F_{\mathcal{B}} \longrightarrow F_{\mathcal{A}}$$

is a positive surjective isometry. Notice that finite trace makes $F_{\mathcal{A}} = L^1(\mathcal{A}, \tau)$ and $F_{\mathcal{B}} = L^1(\mathcal{B}, \nu)$ as sets, though not with their ordinary L^1 norms.

Step 4: apply the sum-space classification and return. Both total traces are now greater than one, so Theorem 4.2 and its positive corollary in the source paper apply to V [1, Theorem 4.2]. They show that V is the restriction of a trace-preserving Jordan $*$ -isomorphism

$$J : \mathcal{B} \longrightarrow \mathcal{A}.$$

We finally identify the original map. Jordan isomorphisms preserve $a \circ b = (ab + ba)/2$, and a trace satisfies $\mu(ab) = \mu(a \circ b)$. Therefore, for $b \in \mathcal{B}$ and $x \in \mathcal{A}$,

$$\tau(J(b)x) = \tau(J(b) \circ x) = \tau(J(b \circ J^{-1}(x))) = \nu(b \circ J^{-1}(x)) = \nu(bJ^{-1}(x)). \quad (3)$$

Equations (1), (3), and $V = J$ give $\nu(bU(x)) = \nu(bJ^{-1}(x))$ for all $b \in \mathcal{B}$. Since bounded normal functionals separate points of the finite algebra $\mathcal{B}$, $U(x) = J^{-1}(x)$ for all $x \in \mathcal{A}$. Hence U is a trace-preserving Jordan $*$ -isomorphism. In particular it is unital, so

$$\tau(\mathbf{1}_{\mathcal{A}}) = \nu(\mathbf{1}_{\mathcal{B}}).$$

The converse is exactly the norm calculation in the source theorem: a trace-preserving Jordan $*$ -isomorphism preserves both the L^1 norm and the operator norm, hence their maximum. This completes the proof. $\square$

4 Verification, limitations, and novelty status

The proof is non-computational. Its two external mathematical inputs were checked in the locally stored papers: the order-isomorphism/normality result is Lemma 3.1 of [2], and the sum-space classification is Theorem 4.2 with its positive corollary in [1]. The source paper’s Proposition 5.1 supplies the extreme-point characterization used only to exclude a target total at most one.

The theorem addresses positive surjective linear isometries under the source paper’s standing non-atomicity and semifiniteness assumptions. It does not classify non-positive isometries, nor does it remove non-atomicity.

A bounded literature search through 27 August 2026 used arXiv:2007.02324, the exact title and authors, the quoted open-problem phrase, combinations of “positive surjective isometry”, “ $L^1 \cap L^\infty$ ”, and “trace between 1 and 2”, and a forward-citation check. It found the arXiv/journal versions of the source and one unrelated citing paper, but no later claimed answer to this question. This supports, but does not guarantee, novelty.

Human-review recommendation. Review as a candidate full result. The main checkpoint is formula (2) on the normal predual and its use to make the preadjoint an onto sum-space isometry. The order-unit and trace-pairing arguments should be checked next.

References

- [1] P. de Jager and J. Conradie, *Extreme point methods in the study of isometries on certain non-commutative spaces*, arXiv:2007.02324; Glasgow Math. J., doi:10.1017/S0017089521000227.
- [2] P. de Jager and J. J. Conradie, *Isometries between non-commutative symmetric spaces associated with semi-finite von Neumann algebras*, Positivity **24** (2020), 815–835; arXiv:1907.06452.
- [3] P. G. Dodds, T. K.-Y. Dodds, and B. de Pagter, *Non-commutative Köthe duality*, Trans. Amer. Math. Soc. **339** (1993), 717–750.